

Received 24 October 2023, accepted 15 November 2023, date of publication 20 November 2023,
date of current version 29 November 2023.

Digital Object Identifier 10.1109/ACCESS.2023.3334622

RESEARCH ARTICLE

Vehicle Trajectory Prediction Based on Multivariate Interaction Modeling

DONGXIAN SUN¹, HONGWEI GUO¹, AND WUHONG WANG

School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China

Corresponding author: Hongwei Guo (guohongwei@bit.edu.cn)

This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFB3301605, and in part by the National Natural Science Foundation of China under Grant 52202503 and Grant 52102461.

ABSTRACT Vehicle trajectory prediction is crucial for the safe driving of the intelligent and connected vehicle, but the existing researches suffer from inadequate interaction modeling, especially lacking modeling for dynamic interaction. To solve this problem, we propose a vehicle trajectory prediction model SDT-ATT based on multivariate interaction modeling. Firstly, a hierarchical extracting module is constructed based on the bi-directional long short-term memory network (Bi-LSTM) to extract the contexts of the target vehicle parameters, neighbor vehicle spatial parameters and neighbor vehicle dynamic parameters, respectively. Secondly, the temporal, spatial and dynamic interactions of vehicles are modeled and learned based on the multi-head attention (MHA) mechanism, then fused to obtain the multivariate interaction features. Next, a direct multi-step prediction module is constructed based on long short-term memory network (LSTM), which directly outputs the future coordinates of the vehicle in the target prediction horizons. Finally, the vehicle trajectory prediction model SDT-ATT proposed in this paper is validated based on the public naturalistic driving datasets I-80 and US-101. The data results show that compared with the CS-LSTM model, the SDT-ATT has 9.3% lower ADE average value, 8.5% lower FDE average value and 8.7% lower RMSE average value, and the inference speed of the SDT-ATT model meets the real-time requirement.

INDEX TERMS Intelligent and connected vehicle, vehicle trajectory prediction, multivariate interaction modeling, multi-head attention mechanism.

I. INTRODUCTION

Intelligent and connected vehicle (ICV) is equipped with cameras, lidar and other on-board sensors as well as controllers, actuators and other devices, integrating modern network communication technology, which can realize the functions of environment perception, intelligent decision-making and cooperative control [1]. In the traditional vehicle driving process, he/she observes the historical motion states of the surrounding vehicles, reasons about vehicles' future trajectories, and estimates the potential driving risks of the ego-vehicle, to perform the corresponding driving operations. In autonomous driving, the trajectory prediction module serves as a bridge between environment sensing and decision making, and the efficiency and effectiveness of the

trajectory prediction directly affect the rationality and safety of the ICV's subsequent decision making and path planning [2]. BMW predicts driving intentions of surrounding vehicles based on Dynamic Bayesian Networks to improve the driving safety [3]. And Baidu Apollo autonomous vehicle uses Inter-TNT model as the prediction module [4]. However, due to the complex interaction behaviors between vehicles, how to accurately and efficiently predict and output the vehicle trajectories in the future has become a recognized problem and challenge in academia and industry.

Existing vehicle trajectory prediction methods can be divided into four categories [5]: physics-based methods, classic machine learning-based methods, deep learning-based methods, and reinforcement learning-based methods. Earlier, scholars mostly predicted the future motion state of vehicles based on physical rules, and such methods mainly include the Constant-Velocity (CV) model [6], Constant-Accelerate

The associate editor coordinating the review of this manuscript and approving it for publication was Aysegül Ucar¹.

(CA) model [7] and Kalman Filtering (KF) based methods [8], Monte Carlo (MC) based methods [9]. However, this category is only applicable to short-term trajectory prediction with a prediction horizon of no more than 1s, and cannot effectively complete the medium- and long-term prediction tasks. With the development of big data technology and the rapid growth of data, scholars in this field have gradually adopted classical machine learning to enhance the effect of trajectory prediction, mainly including methods based on Gaussian Process (GP) [10], Support Vector Machine (SVM) [11] and Hidden Markov Model (HMM) [12], [13]. However, classic machine learning-based methods are mostly only applicable to simpler driving scenes, and have poor trajectory prediction performances in complex driving scenes. In recent years, the rapid development of reinforcement learning technology has provided new solution ideas for vehicle trajectory prediction [14], [15], but the vehicle trajectory prediction methods based on reinforcement learning require higher computational and time costs, compared to the other prediction methods.

At present, deep learning is the mainstream method for vehicle trajectory prediction [16], [17]. Since trajectory prediction is essentially a time series forecasting problem, and recurrent neural network (RNN) and its variants, long and short-term memory network (LSTM) and gated recurrent unit (GRU), can effectively extract the long-range temporal dependencies, this type of network is applied to the vehicle trajectory prediction task. J et al. [19] proposed vehicle trajectory prediction based on intention recognition enhancement by LSTM, which effectively improved the prediction accuracy. Min et al. [20] processed the camera, lidar, and GPS time series data collected by the naturalistic driving study based on RNN, to predict the trajectory of the obstacle vehicle. Convolutional neural network (CNN) has achieved great success in the field of computer vision. Affected by this, scholars have tried to apply it to the field of trajectory prediction and achieved good results. Deo et al. [21] introduced the convolutional layers to replace the fully connected layers in S-LSTM [22], which improved the modeling ability of spatial interaction between vehicles and effectively improved the trajectory prediction effect of the target vehicle. The emergence of transformer (TF) [23] has triggered scholars to pay attention to attention mechanism (AM) [24], [25], [26], [27]. Jin et al. [28] used the attention mechanism to estimate the driving behavior of the agent, thereby enhancing the trajectory performances.

Scholars in the field have invested great efforts in vehicle trajectory prediction and have achieved some results. But there is still the problem of insufficient vehicle interaction modeling. In terms of temporal interaction, study either extracted temporal dependencies based on RNN [29], or based on the hidden features extracted by the spatial interaction module to extract temporal dependencies [30]. However, the former is relatively simple and dose not effectively model the temporal interaction; the latter destroys the original temporal interaction structure to a certain extent.

In recent years, scholars have paid more and more attention to the spatial interaction between vehicles, and proposed a variety of spatial interaction modeling methods based on vehicle coordinate data [22], [31], [32], but ignored the importance of inter-vehicle dynamic interaction for the future motion trajectory of the target vehicle.

As shown in Fig. 1a, at this moment, the euclidean distance between the gray vehicle and the red vehicle is a constant value, i.e., the intensity of spatial interaction is a constant value, but the different driving velocities of the red vehicle will lead to different driving behaviors of the gray vehicle: when the red vehicle is driving straight at a higher velocity, the gray vehicle will usually choose to pause the lane change to avoid the potential collision risk and execute the lane change after the red vehicle is far away, as shown in Fig. 1b; when the red vehicle is driving at a slower velocity, the gray vehicle will execute the lane change directly to the left, as shown in Fig. 1c. In addition, the different motion states of the neighbor vehicle will lead to different future trajectory of the target vehicle in Fig. 1d: when the red vehicle accelerates, the gray vehicle will keep the current driving state or even accelerate, as shown in Fig. 1e; when the red vehicle slows down, to avoid a collision, the gray vehicle will choose to slow down, as shown in Fig. 1f.

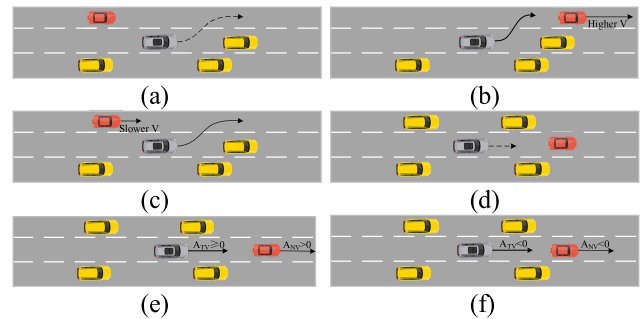


FIGURE 1. An illustration of the dynamic interaction, where the black dotted line indicates the driving intention of the vehicle and the black solid line indicates the motion trajectory of the vehicle. V indicates the velocity of the vehicle, and A indicates the acceleration of the vehicle. (a) The gray vehicle in the middle lane will change lanes to the left in search of a better driving experience, and there is a red neighbor vehicle going straight behind the target lane. (d) The gray vehicle in the middle lane is going straight following the red front vehicle.

To address the above problems and model multivariate interaction between vehicles, this paper proposes a trajectory prediction scheme for ICVs, namely SDT-ATT ('S' denotes spatial interaction, 'D' denotes dynamic interaction, 'T' denotes temporal interaction, and 'ATT' denotes the attention mechanism used in our modeling method). Firstly, the hierarchical extraction module based on Bi-LSTM is constructed to extract the context information. Then, the multivariate interaction module is constructed based on the multi-headed attention mechanism to effectively model the multivariate interaction between vehicles. Finally, the Direct Multi-Step (DMS) prediction module is constructed based on LSTM to output the future trajectory of the target vehicle.

The main contributions of our research are as follows.

- Multivariate interaction modeling is proposed to describe and learn the interaction behaviors between vehicles, and our comparative analysis and ablation study demonstrate this method can improve the accuracy of vehicle trajectory prediction.
- A direct multi-step prediction module is proposed, and it outputs the prediction results in one forward step rather than a step-by-step way, which makes the real-time prediction of the target vehicle possible.

The rest of this paper is arranged as follows. Section II describes the research problem of this paper. Section III introduces the SDT-ATT model in detail. Section IV presents the experiment results and discusses the trajectory prediction performance of SDT-ATT. Section V concludes this paper.

II. PROBLEM DESCRIPTION

Real-time and accurate trajectory prediction results can help ICVs understand the current driving scene, perceive driving risks in advance, and make effective decisions and plans. In this paper, we assumed that the position coordinates information of vehicles in the scene has been obtained through the intelligent network environment. We denote the coordinates information X of vehicles in the scene over a past time horizon t_h as:

$$X = [P_1, P_2, \dots, P_{t_h}] \quad (1)$$

$$P_t = [(x_t^e, y_t^e), (x_t^1, y_t^1), (x_t^2, y_t^2), \dots, (x_t^N, y_t^N)] \quad (2)$$

where P_t is coordinate information of all vehicles in the scene at the time t , e indicates the target vehicle, N is the number of the neighbor vehicles, and (x, y) is the position coordinate of the vehicle. We denote the predicted trajectory of the target vehicle in the future time horizon t_f as:

$$Y = [\hat{P}_{t_h+1}, \hat{P}_{t_h+2}, \dots, \hat{P}_{t_h+t_f}] \quad (3)$$

$$\hat{P}_{t_h+t'} = (\hat{x}_{t_h+t'}^e, \hat{y}_{t_h+t'}^e) \quad (4)$$

where Y is the predicted trajectory of the target vehicle, $\hat{P}_{t_h+t'}$ is the predicted position coordinate at the time $t_h + t'$.

Therefore, in this research, the vehicle trajectory prediction problem can be summarized as follows: the historical trajectory information of the target vehicle and its surrounding neighbor vehicles is input into the vehicle trajectory prediction model, and the multivariate interactions between vehicles are modeled by the learning ability of the neural network model itself, to output the motion trajectory of the target vehicle in the future period. The illustration of the research scene is shown in Fig. 2.

III. THE SDT-ATT MODEL

Vehicle trajectory prediction not only needs to consider the historical trajectory information of the predicted vehicle, but also the complex interaction behaviors between the predicted vehicle and neighbor vehicles. Therefore, in this paper, comprehensively considering temporal, spatial and

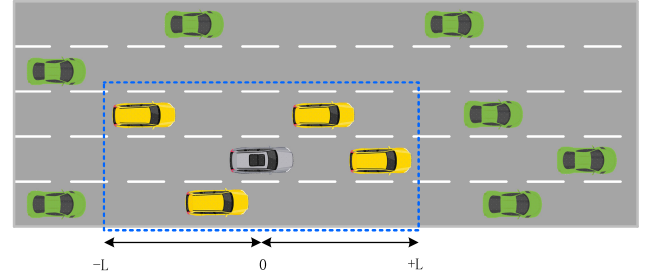


FIGURE 2. An illustration of the research scene. The green vehicles indicate the unrelated vehicles that have no spatial and dynamic interaction with the target vehicle. And the blue dashed box indicates the interaction range of the target vehicle.

dynamic interactions, a vehicle trajectory prediction model, SDT-ATT, is proposed. The framework of the model is shown in Fig. 3, and the SDT-ATT model mainly includes three parts: 1) the hierarchical extracting module, based on Bi-LSTM, respectively extracts the context information of the target vehicle parameters, NV spatial parameters, and NV dynamic parameters; 2) the multivariate interaction module, based on the multi-head self-attention mechanism, extracts temporal, spatial and dynamic interaction features hidden in context information, then fuses them to obtain hidden features of multivariate interaction; 3) the direct multi-step trajectory prediction module, based on the LSTM network decode the multivariate interaction, directly outputs the trajectory of the target vehicle in the future.

A. HIERARCHICAL EXTRACTING BASED ON Bi-LSTM

The hierarchical extracting module consists of 3 embedding layers and multiple Bi-LSTM layers. Specifically, the embedding layers are composed of three parallel fully connected layers (FC), respectively processing TV parameters (position coordinates, velocity and acceleration), NV spatial parameters (position coordinates) and NV dynamic parameters (velocity and acceleration), and the activation function is LeakyReLU. It is worth noting that the velocity and acceleration information in the TV parameters and NV dynamic parameters are obtained by approximate calculation with the following formulas:

$$\left. \begin{aligned} V_{x,t} &= \frac{x_{t+1} - x_t}{1/f}, V_{y,t} = \frac{y_{t+1} - y_t}{1/f} \\ A_{x,t} &= \frac{V_{x,t+1} - V_{x,t}}{1/f}, A_{y,t} = \frac{V_{y,t+1} - V_{y,t}}{1/f} \end{aligned} \right\} \quad (5)$$

where $V_{x,t}$ and $V_{y,t}$ are the lateral and longitudinal velocity of the vehicle at t time moment, $A_{x,t}$ and $A_{y,t}$ are the lateral and longitudinal acceleration of the vehicle at t time moment, and f is the sampling frequency of the data.

RNN is effective in extracting valid information from time series data, but due to itself structure, it cannot handle long-term dependencies and is prone to gradient disappearance [33]. As a variant of RNN, LSTM introduces a gating mechanism (input gate, forget gate and output gate) to control the transmission of information, which can effectively avoid the problem of gradient disappearance. Further, the Bi-LSTM

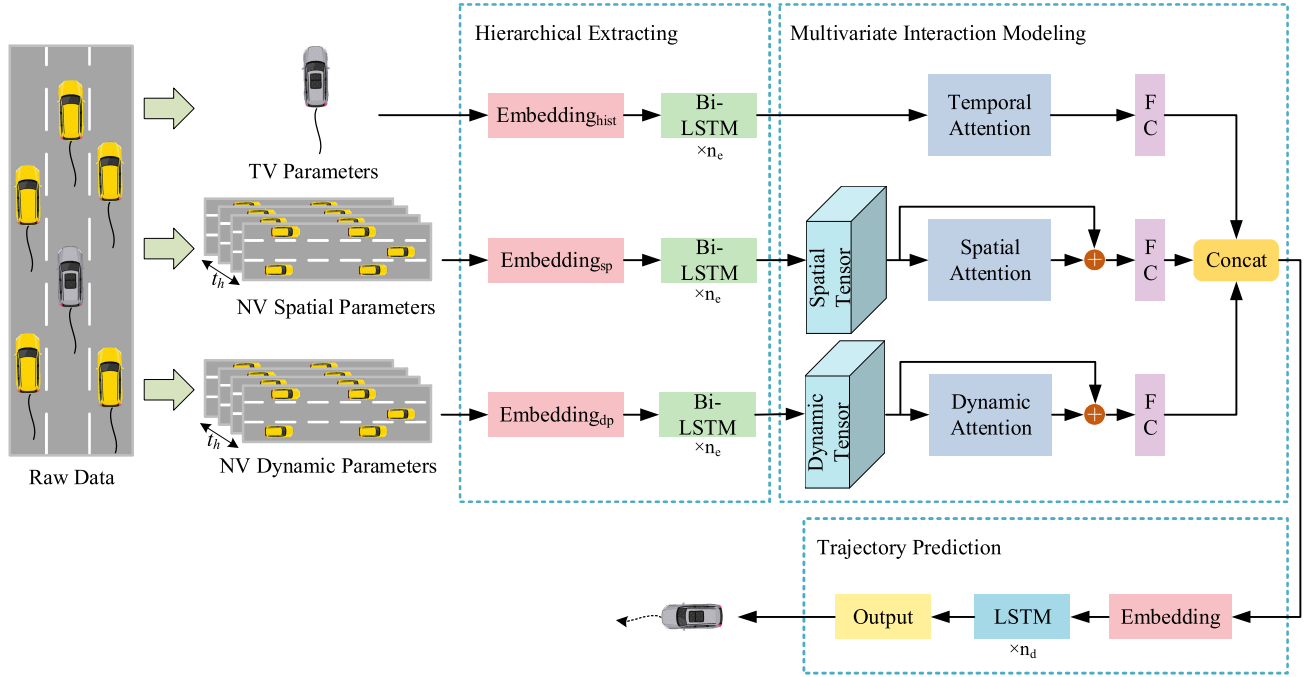


FIGURE 3. The framework of SDT-ATT model. TV indicates the target vehicle, NV indicates the neighbor vehicle. TV parameters include position coordinates and velocity and acceleration obtained through coordinate calculation. NV spatial parameters only include position coordinates of the neighbor vehicles, and NV dynamic parameters only include velocity and acceleration of the neighbor vehicles obtained through coordinate calculation.

adds a reverse layer based on LSTM, which enables the Bi-LSTM to look at a particular sequence both from front to back as well as from back to front and helps to improve the performance of the prediction model. The structure of Bi-LSTM is shown in Fig. 4. Considering the above factors, in this paper, Bi-LSTM layers are selected to form the hierarchical extracting module, and the feature information processed by the embedding layer is input into it to extract the hidden features in the time series data:

$$\begin{cases} (h_t^{fw}, c_t^{fw}) = \text{LSTM}_{fw}(e_t, h_{t-1}, c_{t-1}; W_{fw}) \\ (h_{t_h-t+1}^{bw}, c_{t_h-t+1}^{bw}) = \text{LSTM}_{bw}(e_{t_h-t+1}, \\ h_{t_h-t+2}, c_{t_h-t+2}; W_{bw}) \\ h_t = \text{concat}(h_t^{fw}, h_{t_h-t+1}^{bw}) \end{cases} \quad (6)$$

$$\begin{cases} H_{hist} = (h_1^{hist}, h_2^{hist}, \dots, h_{t_h}^{hist}) \\ H_{nbrs_l} = h_{t_h}^{nbrs_l} \\ H_{nbrs_d} = h_{t_h}^{nbrs_d} \end{cases} \quad (7)$$

where LSTM_{fw} and LSTM_{bw} represent all calculations of the forward and reverse LSTM unites in Bi-LSTM, respectively, e_t is the input at time t , h_t and c_t are the hidden state and memory cell internal state, h_t^{fw} is the output of the forward LSTM unit at time t , $h_{t_h-t+1}^{bw}$ is the output of the reverse LSTM unit at time $t_h - t + 1$, W_{fw} and W_{bw} are the learnable parameters, H_{hist} , H_{nbrs_l} , H_{nbrs_d} are the extracting features of TV parameters, NV spatial parameters and NV dynamic parameters, i.e., the final output of the hierarchical extraction module. It is worth mentioning that the parameters are shared between Bi-LSTM, as shown in Fig. 3.

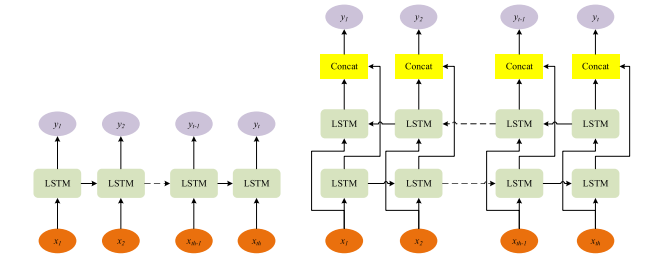


FIGURE 4. Architecture comparison of LSTM network (left) and Bi-LSTM network we used (right). Compared to LSTM, the presence of a reverse layer allows Bi-LSTM to fully extract contextual information, e.g., the output y_2 not only contain the information of x_1 and x_2 , but also x_3 to x_{th} .

B. MULTIVARIATE INTERACTION MODELING BASED ON MULTI-HEAD ATTENTION

1) TEMPORAL INTERACTION

The historical states of the vehicle at different moments have different influences on its future motion [34], and how to effectively model this temporal interaction is crucial to the accuracy of trajectory prediction. The attention mechanism can automatically capture and learn the weight of the vehicle historical state at different moments, i.e., the strength of temporal interaction. Multi-head attention allows the model to focus on information from different locations at the same time [23], which helps to increase the stability and robustness of the model [35]. Therefore, in this paper, we choose the multi-head attention mechanism to build the temporal attention module, so as to model and learn the temporal interaction behavior of vehicles. First, H_{hist}

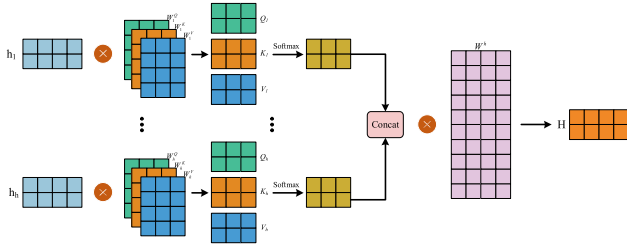


FIGURE 5. The architectures of multi-head attention, where Q , K , and V indicate the query matrix, key matrix, and value matrix.

extracted from the TV parameters is input into the temporal attention module to model the temporal dependency, and then the feature H_T is input into the FC layer for nonlinear transformation to obtain the temporal interaction information H_{TI} . The specific calculation process for obtaining H_{TI} is as follows.

$$\begin{cases} Q_{TI_i} = \phi(H_{hist_i}, W_i^Q) \\ K_{TI_i} = \phi(H_{hist_i}, W_i^K) \\ V_{TI_i} = \phi(H_{hist_i}, W_i^V) \end{cases} \quad (8)$$

$$head_i = \text{softmax}\left(\frac{Q_{TI_i} K_{TI_i}^T}{\sqrt{d_K}}\right) V_{TI_i} \quad (9)$$

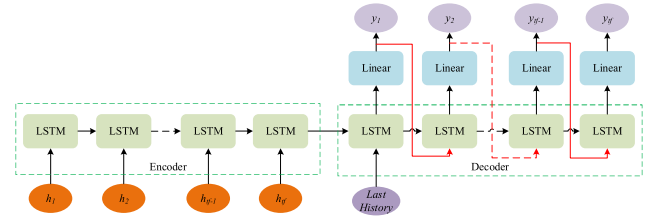
$$H_T = \text{Concat}(head_1, head_2, \dots, head_h) W^H \quad (10)$$

$$H_{TI} = f(\phi(H_T, W_T)) \quad (11)$$

where H_{hist_i} is the subsequence of H_{hist} , W_i^Q , W_i^K , W_i^V , W^H and W_T are trainable parameters, Q_{TI_i} , K_{TI_i} , V_{TI_i} are the query matrix, key matrix, and value matrix of scaled dot-product attention respectively, d_K is the scaling factor, $head_i$ is the output of the i -th attention head, H_{TI} is the temporal interaction feature. The structures of multi-head attention are shown in Fig. 5.

2) SPATIAL AND DYNAMIC INTERACTION

In recent years, scholars have paid more and more attention to the spatial interaction between vehicles in driving scene and proposed various spatial interaction modeling methods [36], [37], [38], but neglected the importance of dynamic interaction modeling for vehicle motion. In this paper, the method of combining social tensor and multi-head attention is used to extract and model the spatial interaction features and dynamic interaction features of neighbor vehicles. The social tensor was first proposed in the S-LSTM model to extract the spatial interaction behavior between pedestrians, and then it was applied to the field of vehicle trajectory prediction after being adjusted and optimized in the CS-LSTM model that has now become one of the comparison baselines for vehicle trajectory prediction. Inspired by CS-LSTM, we firstly construct spatial tensor and dynamic tensor based on H_{nbrs_l} and H_{nbrs_d} obtained by hierarchical extracting. The constructions of spatial tensor H_{spa_tensor} and dynamic tensor H_{dyn_tensor} are based on the following



(a) Prediction method based on Seq2Seq

(b) The prediction method we use

FIGURE 6. Comparison of direct multi-step and iterative multi-step prediction networks.

formulas:

$$\begin{cases} H_{spa_tensor} = \sum_{i \in N_e} I(X^i, X^e) H_{nbrs_l} \\ H_{dyn_tensor} = \sum_{i \in N_e} I(X^i, X^e) H_{nbrs_d} \end{cases} \quad (12)$$

where X^i is the coordinate matrix of neighbor vehicle i , X^e is the coordinate matrix of target vehicle, $I()$ is the indicator function proposed in S-LSTM [22], N_e is the set of neighbor vehicles of target vehicle. Then we input two tensors separately into spatial attention mechanism and dynamic attention module based on multi-head attention to extract interaction features H_S and H_D hidden in tensors, similar to the operation of temporal interaction modeling. At the same time, to avoid the vanishing gradient problem, the residual connection [39] is introduced. Therefore, the final spatial and dynamic interaction feature are obtaining by following formulas:

$$H_{SI} = f(\phi(H_S, W_S)) + H_{spa_tensor} \quad (13)$$

$$H_{DI} = f(\phi(H_D, W_D)) + H_{dyn_tensor} \quad (14)$$

where W_S and W_D are learnable parameters.

The output of the multivariate interaction modeling module is the fusion of temporal, spatial and dynamic interaction feature, i.e., the multivariate interaction feature H_{MVI} between vehicles:

$$H_{MVI} = \text{Concat}(H_{TI}, H_{SI}, H_{DI}) \quad (15)$$

C. TRAJECTORY PREDICTION BASED ON LSTM

Currently, scholars mostly obtain the final predicted output based on Seq2Seq, the structure of which is shown in Fig. 6a. In Seq2Seq structure, the output of the previous prediction time step is input into the current prediction time step together with the hidden state, and so on until all the predicted outputs in the target prediction time are obtained. We believe that the above iterative loop not only increases the inference time of the model, but also exacerbates the prediction error accumulation of LSTM by inputting the outputs of the previous

prediction time step and the hidden state into the current prediction time step. Therefore, to avoid the problem of error accumulation caused by iterative multi-step prediction and to improve the inference speed of the prediction model [40], we build a direct multi-step prediction module based on LSTM as shown in Fig. 6b, which directly (one forward step) outputs the trajectory of the target vehicle in the prediction horizons instead of reasoning through step-by-step iterations. Trajectory prediction module mainly consists of embedding layer, LSTM layer and output layer. The embedding layer is used to process the time dimension of multivariate interaction features and convert it into the length of the target prediction horizon t_f . The predicted output is calculated as:

$$H_{de_emb} = f(\phi(H_{MVI}, W_{de_emb})) \quad (16)$$

$$H_{de, _} = \text{LSTM}(H_{de_emb}, W_{de}) \quad (17)$$

$$OP = \phi(H_{de}, W_{op}) \quad (18)$$

where W_{de_emb} , W_{de} and W_{op} are the learnable parameters, $\text{LSTM}()$ represents all calculations of the LSTM network, H_{de_emb} is the embedded feature information of the embedding layer, H_{de} is the output of the LSTM network, ‘ $_$ ’ is the hidden state and memory cell internal state of the LSTM network, which is not used and therefore simplified to represent, OP is the output of the model.

Similar to the previous research [15], [16], this research assumes that the future trajectory coordinates of the target vehicle predicted by the SDT-ATT obey the bivariable Gaussian distribution:

$$(\hat{x}_{t_h+t'}, \hat{y}_{t_h+t'}) \sim \mathcal{N}(\hat{\mu}_{t_h+t'}, \hat{\sigma}_{t_h+t'}, \hat{\rho}_{t_h+t'}) \quad (19)$$

where $\hat{\mu}_{t_h+t'} = (\hat{\mu}_x, \hat{\mu}_y)_{t_h+t'}$ is the mean, $\hat{\sigma}_{t_h+t'} = (\hat{\sigma}_x, \hat{\sigma}_y)_{t_h+t'}$ is the standard deviation, and $\hat{\rho}_{t_h+t'}$ is the correlation coefficient.

Finally, we train SDT-ATT and update the model parameters by minimizing the negative log-likelihood loss:

$$\text{Loss}(W) = - \sum_{t_p=t_h+1}^{t_h+t_f} \log(\mathbb{P}(x_{t_p}, y_{t_p} | \hat{\mu}_{t_p}, \hat{\sigma}_{t_p}, \hat{\rho}_{t_p})) \quad (20)$$

where W denotes all trainable parameters in the model.

IV. EXPERIMENTAL EVALUATION AND ANALYSIS

A. EVALUATION DATASETS

To evaluate the vehicle trajectory prediction model SDT-ATT based on multivariate interaction proposed in this paper, we verified and analyzed the prediction effectiveness and efficiency of the model on the I-80 and US-101 datasets in NGSIM [41]. Each dataset contains 45-minute trajectories of real freeway traffic. The data sampling frequency is 10Hz, and the collected parameters include vehicle coordinates, velocity, acceleration, etc. Datasets contain rich scene (mild, moderate and heavy traffic scenes) information, which helps to validate the effectiveness and robustness of the trajectory prediction model.

For the fairness and effectiveness of the comparison experiments, we follow the data processing method of Deo et al. [21]: the raw data are down-sampled from 10Hz to 5Hz, and then each trajectory is divided into segments of 8 seconds, in which the data of the first 3 seconds is used as the historical trajectory to extract multivariate interaction information, and the data of the last 5 seconds is used as the ground truth to calculate the error between ground truth and the predicted trajectory.

B. EVALUATION METRICS

In order to achieve quantitative evaluation, the average displacement error (ADE), final displacement error (FDE), root mean squared error (RMSE), and miss rate (MR) are used to evaluate the performance of the trajectory prediction model. MR is the ratio of cases where the prediction trajectory is not within 2.0 meters of the ground truth based on the L2 distance of the final position [5]. The other three metrics are calculated as:

$$\text{ADE} = \frac{1}{N} \sum_{i=1}^N \sqrt{(\hat{x}^i - x^i)^2 + (\hat{y}^i - y^i)^2} \quad (21)$$

$$\text{FDE} = \sqrt{(\hat{x}_{t+t_f} - x_{t+t_f})^2 + (\hat{y}_{t+t_f} - y_{t+t_f})^2} \quad (22)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N [(\hat{x}^i - x^i)^2 + (\hat{y}^i - y^i)^2]} \quad (23)$$

where x^i and y^i are the true position coordinates of the target vehicle, $\hat{x}^i$ and $\hat{y}^i$ are the predicted position coordinates, $t+t_f$ is the final prediction time step, N is the number of samples. The smaller the values of the evaluation metrics, the better the performance of the prediction model.

C. IMPLEMENTATION DETAILS

The vehicle trajectory prediction model proposed in this paper is implemented based on the Python programming language and the PyTorch deep learning framework. The implementation details are as follows.

- 1) Spatial interaction range: Similar to CS-LSTM, this research restricts the spatial interaction range to two adjacent lanes of the target vehicle in the lateral direction and ± 90 feet in the longitudinal direction.
- 2) Embedding layers: In the hierarchical extracting module, three fully connected layers act on the feature dimensions of the data, expanding the original feature dimensions to 32. The increase in dimensionality is beneficial to improve the learning ability of the prediction model. In the prediction module, an embedding layer is used to dilate the temporal dimension to the length of the prediction horizons, i.e., 25.
- 3) Number of LSTM layers: To obtain the best trajectory prediction effect, set the number of layers of Bi-LSTM in the hierarchical extracting module and LSTM in the prediction module to different numbers, train the model. The prediction effects of the model under

different network layers are shown in Table 1, where the first column indicates the number of layers of Bi-LSTM n_e in the hierarchical extracting module, and the first row indicates the number of layers of LSTM n_d in the prediction module. The three values are ADE, FDE and RMSE. Therefore, we determine the final network configuration as $n_e = 4$ and $n_d = 1$.

- 4) Multi-head self-attention: The number of heads of temporal attention is set to be 2, and the number of heads for spatial and dynamic attention is set to be 3.
- 5) Training process: The network parameters are trained based on NVIDIA GTX3080 GPU. The batch size is set to 128, and we use Adam optimizer with 0.001 learning rate. The number of hidden units in Bi-LSTM is 64, and in LSTM is 128, using the leaky_ReLU activation function.

TABLE 1. Comparison of prediction errors for the model with different number of LSTM layers. Data are in meters.

$n_d \backslash n_e$	1	2	3
1	0.70/1.55/2.13	0.70/1.56/2.15	0.69/1.55/2.14
2	0.69/1.54/2.11	0.69/1.54/2.13	0.70/1.57/2.16
3	0.68 /1.52/2.10	0.70/1.55/2.13	0.70/1.56/2.15
4	0.68 / 1.51 / 2.09	0.70/1.56/2.15	0.70/1.56/2.14
5	0.69/1.53/2.12	0.70/1.55/2.14	0.70/1.57/2.17

D. QUANTITATIVE ANALYSIS

1) QUANTITATIVE ANALYSIS OF PREDICTION ERRORS

To validate the effectiveness of the prediction model proposed in this paper, the following models were selected as comparison baselines.

- Constant Velocity (CV): This method uses a constant velocity Kalman filter to predict the future trajectory of the target vehicle, which does not consider the interaction between vehicles.
- C-VGMM+VIM: This method uses hidden Markov models to estimate driving intentions, then predict the trajectory based on variational Gaussian mixture models. This baseline only considers the spatial interaction between vehicles.
- V-LSTM: This baseline learns the time series information hidden in the historical trajectory data, then outputs the trajectory distribution of the target vehicle based on the LSTM encoder-decoder network. V-LSTM does not consider the interdependencies between vehicles.
- Bi-GRU: Compared to V-LSTM, the LSTM layer in the encoder is replaced with Bi-GRU to make full use of the context information of the vehicle historical trajectory, and the LSTM layer in the decoder is replaced with GRU. This method does not consider the interdependencies between vehicles.
- Bi-LSTM: Compared to V-LSTM, the LSTM layer in the encoder is replaced with Bi-LSTM to make full

use of the context information of the vehicle historical trajectory. The rest of the settings are the same. This method does not consider the interdependencies between vehicles.

- S-LSTM: This baseline proposed in the field of pedestrian trajectory prediction is transferred to vehicle trajectory prediction, and the spatial grid of the social tensor is modified to 13×3 as described in [21]. The model only considers the spatial interaction behavior of the target vehicle with the neighbor vehicles.
- CS-LSTM: Compared to S-LSTM, the fully connected layers are replaced with the convolutional pooling network. This baseline also only considers spatial interaction.
- SVT-ATT: Compared to the SDT-ATT model, this model removes the vehicle acceleration parameters and simplifies the dynamic interaction modeling.

In the field of vehicle trajectory prediction, scholars mostly use RMSE to evaluate the performance of the model, so in this paper, we also choose it as one of the evaluation metrics. We compare the RMSE values of our model with the baselines on I-80 and US-101 datasets, as shown in Table 2. We can see that the RMSE values of CV and C-VGMM+VIM are similar to the RESE values of the simple neural network V-LSTM, but much larger than those based on the deep learning methods. Compared with S-LSTM and CS-LSTM, which only consider spatial interactions, the methods proposed in this paper integrate the multivariate interactions between vehicles, and the prediction errors are further reduced. Specifically, compared to CS-LSTM, the average REMSE values are reduced by 6.6% for SVT-ATT and 8.7% for SDT-ATT. In addition, we also compare the prediction errors of our model with the above baselines on ADE, FDE, and MR values, as shown in Table 3. We can see that the error values of the Bi-GRU and Bi-LSTM are smaller than those of V-LSTM for the same prediction horizons without considering vehicle interaction, which indicates the importance of making full use of the context information for vehicle trajectory prediction. Meanwhile, S-LSTM and CS-LSTM introduce spatial interaction modeling to improve the accuracy of trajectory prediction. Our SVT-ATT model, which simultaneously models temporal, spatial, and dynamic interactions, further improves the trajectory prediction accuracy and fully demonstrates the importance and necessity of multivariate interaction modeling for vehicle trajectory prediction. The comparison of prediction error data between SVT-ATT and SDT-ATT shows that acceleration contributes to the completeness of the motion interaction modeling. What's more, our SDT-ATT model outperforms other baselines in all prediction horizons, and the three metrics at each prediction moment are the minimum values. Specifically, compared with the CS-LSTM, the average ADE values of SDT-ATT are 9.3% lower, and the average FDE values are 8.5% lower, which effectively demonstrates the prediction accuracy of our model.

TABLE 2. Comparison of RMSE for different model. The Performance of models (denoted by *) is published in [16].

Metrics	Prediction Horizons (s)	Compared baselines					Our models	
		CV*	C-VGMM+VIM*	V-LSTM*	S-LSTM*	CS-LSTM*	SVT-ATT	SDT-ATT
RMSE (m)	1	0.73	0.66	0.68	0.65	0.61	0.55	0.52
	2	1.78	1.56	1.65	1.31	1.27	1.17	1.14
	3	3.13	2.75	2.91	2.16	2.09	1.95	1.89
	4	4.78	4.24	4.46	3.25	3.10	2.91	2.84
	5	6.68	5.99	6.27	4.55	4.37	4.13	4.04
	Average	3.42	3.04	3.19	2.38	2.29	2.14	2.09

TABLE 3. Comparison of ADE, FDE, and MR for different model.

Metrics	Prediction Horizons (s)	Compared baselines					Our models	
		V-LSTM	Bi-GRU	Bi-LSTM	S-LSTM	CS-LSTM	SVT-ATT	SDT-ATT
ADE (m)	1	0.22	0.22	0.22	0.21	0.21	0.19	0.19
	2	0.53	0.52	0.52	0.45	0.44	0.41	0.40
	3	0.93	0.91	0.91	0.72	0.71	0.66	0.64
	4	1.40	1.38	1.37	1.02	1.01	0.95	0.92
	5	1.94	1.92	1.90	1.38	1.36	1.28	1.24
	Average	1.00	0.99	0.99	0.76	0.75	0.70	0.68
FDE (m)	1	0.42	0.42	0.42	0.38	0.38	0.36	0.33
	2	1.16	1.14	1.14	0.90	0.89	0.84	0.81
	3	2.14	2.11	2.10	1.52	1.50	1.42	1.37
	4	3.35	3.31	3.29	2.30	2.26	2.12	2.07
	5	4.76	4.71	4.68	3.27	3.21	3.02	2.95
	Average	2.37	2.34	2.32	1.67	1.65	1.55	1.51
Miss Rate		0.74	0.74	0.73	0.60	0.59	0.58	0.56

2) QUANTITATIVE ANALYSIS OF MODEL SIZE AND INFERENCE SPEED

Whether the vehicle trajectory prediction algorithm can be deployed to intelligent and connected vehicles requires not only the prediction effectiveness of the model, but also the model size and inference speed of the model to be considered in combination. A small capacity model can ensure that it can run under limited hardware conditions, and a fast inference speed can ensure that the ICVs have enough time to make decisions. Since the prediction effect of LSTM, Bi-GRU, and Bi-LSTM is far from other models, S-LSTM and CS-LSTM models are selected to compare the model size and inference speed with SDT-ATT model, and the data are shown in Table 4.

Compared with CS-LSTM, the SDT-ATT model proposed in this paper has a more complex structure, so the model size is larger, but it is similar to the S-LSTM model capacity. To ensure the fairness of the comparison, the inference speeds of the three models were all calculated on the NVIDIA GTX 3080 GPU. The result was that the longest inference time of SDT-ATT was 0.081 milliseconds, which was slightly higher than the inference time of the two compared models. Taking an ICV traveling at 120km/h as an example, its travel distance in 0.081 ms is only 0.0027m, so the SDT-ATT vehicle trajectory prediction model fully meets the real-time requirements.

TABLE 4. Comparison of the model size and inference speed of different model.

Models	Model Size (K)	Inference Time (ms)
S-LSTM	1380	0.076
CS-LSTM	757	0.079
SDT-ATT	1333	0.081

E. ABLATION STUDY

1) EFFECTIVENESS OF INTERACTION MODELINGS

This research models the multivariate interactions of vehicles based on the multi-headed attention mechanism. To fully demonstrate the necessity and importance of different interaction modeling for trajectory prediction, three variants of the SDT-ATT model are constructed, trained and evaluated based on the same dataset, then compared with the prediction effectiveness of the SDT-ATT model.

Variant 1 is the temporal interaction missing model (denoted as W/O TI), which removes the temporal interaction module based on multi-head attention and directly fuses the extracting features of the target vehicle parameters H_{hist} with the spatial and motion interaction features (H_{SI} , H_{DI}); Variant 2 is the spatial interaction missing model (denoted as W/O SI), which directly fuses the extracting features of the neighboring vehicle spatial parameters H_{nbrs_l} with the

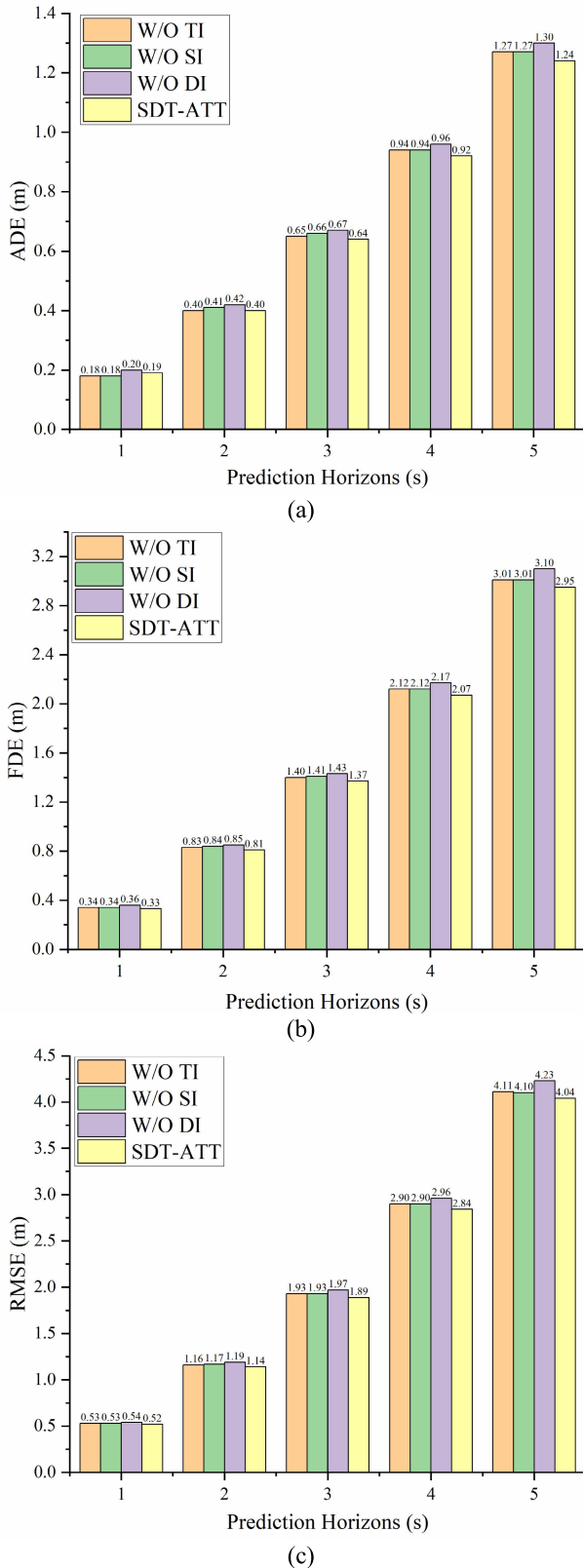


FIGURE 7. Comparison of ADE, FDE and RMSE for the variants of SDT-ATT.

temporal and motion interaction features (H_{TI} , H_{DI}); Variant 3 is the dynamic interaction missing model (denoted as W/O DI), which directly fuses the extracting features of

the neighboring vehicle dynamic parameters H_{nbrs_d} with the temporal and spatial interaction features (H_{TI} , H_{SI}). The prediction errors of the three variant models compared with SDT-ATT are shown in Fig. 7.

The data show that in the prediction horizon of one second, variant 1 and 2 have better performance than SDT-ATT. However, as the prediction horizons increase, the variant models fail to adequately model the interaction behaviors between vehicles, resulting in higher prediction errors than SDT-ATT. According to the histogram shown in Fig. 7, the longer the prediction horizons, the greater the performance gap between the variant model and SDT-ATT, which fully demonstrates the effectiveness and necessity of multivariate interaction modeling proposed in this paper. It is worth noting that the missing temporal interaction and the missing spatial interaction lead to essentially the same error increase values, which indicates that for vehicle trajectory prediction, spatial interaction and temporal interaction modeling are of equal importance. More interestingly, the prediction error increase due to missing dynamic interaction is much larger than the first two variants, which fully indicates that the vehicle dynamic interaction is crucial for vehicle trajectory prediction, and its importance is greater than that of temporal and spatial interactions.

2) SDT-ATT WITH DIFFERENT PREDICTION MODULES

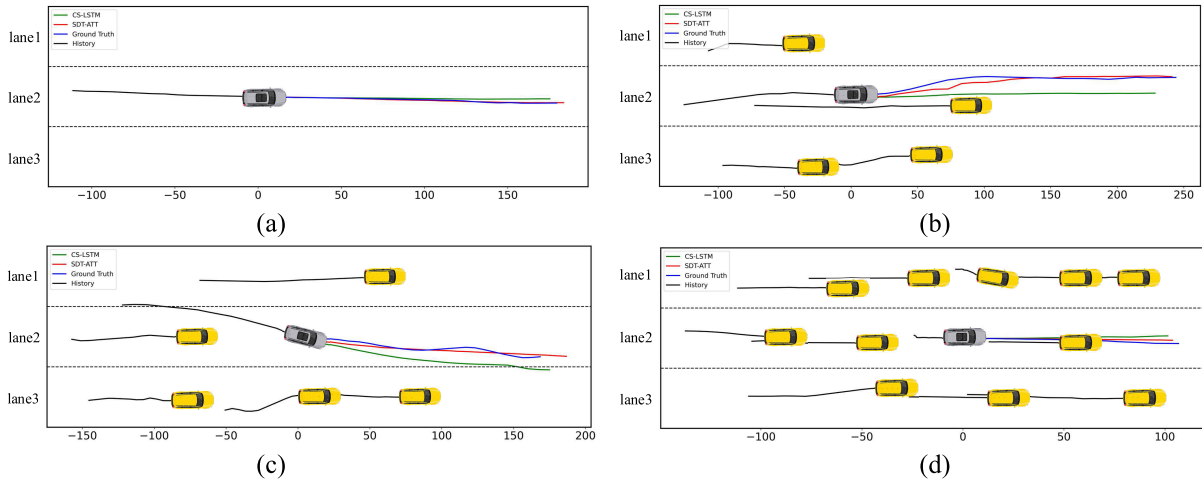
In section III-C, we construct an LSTM-based direct multi-step vehicle trajectory prediction module to inference and output the trajectory of the predicted vehicle at one forward operation rather than a step-by-step way. To verify this method has a positive influence on the prediction effectiveness of the model, we introduce other ways to construct prediction modules as comparisons.

- 1) MLP-based direct multi-step prediction module: This module consists of three fully connected layers. The first fully connected layer acts on the temporal dimension of the input, converting the temporal dimension into the length of the predicted time domain, and the next two fully connected layers act on the feature dimension.
- 2) CNN-based direct multi-step prediction module: We replace the fully connected layers in above 1) with convolutional layers and leave the rest of the settings unchanged.
- 3) Seq2seq-based iterative multi-step prediction module: It consists of two layers of the LSTM network. The first layer is used to encode the input information, and the second layer decodes and predicts the future motion trajectory of the vehicle by iterative inference.

The prediction performances of SDT-ATT with different prediction modules are shown in Table 5. The data shows that the LSTM-based direct multi-step prediction module achieves the best prediction accuracy in all prediction horizons, with the smallest prediction error. Thus, it is clear that the LSTM-based direct multi-step prediction

TABLE 5. Comparison of ADE, FDE and RMSE for the model with different prediction module. Data are converted into meter.

Metrics	Prediction Horizons (s)	Direct Multi-Step			Iterative Multi-Step
		MLP	CNN	LSTM (our)	Seq2Seq
ADE (m)	1	0.22	0.22	0.19	0.30
	2	0.47	0.46	0.40	0.49
	3	0.75	0.74	0.64	0.73
	4	1.06	1.06	0.92	1.01
	5	1.42	1.43	1.24	1.33
	Average	0.78	0.78	0.68	0.77
FDE (m)	1	0.40	0.39	0.33	0.42
	2	0.95	0.94	0.81	0.88
	3	1.58	1.58	1.37	1.45
	4	2.34	2.37	2.07	2.16
	5	3.26	3.32	2.95	3.05
	Average	1.71	1.72	1.51	1.59
RMSE (m)	1	0.58	0.57	0.52	0.59
	2	1.32	1.30	1.14	1.21
	3	2.21	2.20	1.89	1.98
	4	3.28	3.31	2.84	2.95
	5	4.56	4.63	4.04	4.16
	Average	2.39	2.40	2.09	2.18
Inference Time (ms)		0.078	0.076	0.081	0.089

**FIGURE 8.** Visualized prediction results. Black lines are the history trajectories (3 seconds), blue lines are the ground truth (5 seconds), red lines are the predicted trajectories (5 seconds) of our SDT-ATT, and the green lines are the predicted trajectories (5 seconds) of CS-LSTM [21].

module helps to improve the performance of the prediction model. It is worth noting that the inference speed of the direct multi-step class prediction modules is better than that of the iterative multi-step class prediction modules, which is caused by the step-by-step inference structure.

F. QUALITATIVE ANALYSIS

In Fig. 8, we visualize several trajectories predicted by SDT-ATT and CS-LSTM in mild, moderate, and heavy traffic scenes (from a to d) based on I-80 and US-101. Models predict the trajectory of the target vehicle in the next 5 seconds

based on the historical trajectories of the target vehicle and its neighbor vehicles in the past 3 seconds. From Fig. 8, we can notice that:

- From Fig. 8a to 8d, we can clearly see that green lines (CS-LSTM) are farther from the blue lines (ground truth), while red lines (SDT-ATT) can fit blue lines better. This shows that compared with the baseline CS-LSTM, the vehicle trajectory prediction model SDT-ATT proposed in this study can better predict the trajectory of the vehicle in different scenes.
- In Fig. 8b and 8c, the target vehicles perform lane change and overtaking operations for a better driving

experience. And our model accurately predicts these driving intentions and outputs accurate future trajectories, while the prediction results of CS-LSTM deviate from the ground truths.

- Figure 8d shows a relatively heavy traffic scene. At this time, there are 11 adjacent vehicles within the interaction range of the target vehicle, and the interaction between vehicles is very complex. According to the results shown in the Fig. 8d, our model better models the multivariate interaction between vehicles, and the error between the predicted trajectory and the ground truth is smaller than CS-LSTM. This proves that our method is better able to model the multivariate interaction between vehicles.

Based on these observations from the visualized trajectories, we can conclude that compared with the classical CS-LSTM model, the proposed vehicle trajectory prediction model, SDT-ATT, can predict the motion trend of the target vehicle more accurately, and the error between the predicted trajectory and the ground truth is smaller, which has better accuracy and robustness.

V. CONCLUSION

In this paper, a multivariate interaction modeling-based vehicle trajectory prediction model SDT-ATT is proposed to address the problem of inadequate interaction modeling in the field of vehicle trajectory prediction. Based on validation analysis data on publicly available datasets US-101 and I-80, the following main conclusions are drawn.

- In terms of prediction accuracy, the SDT-ATT model proposed in this paper outperforms other comparison baselines in all prediction horizons of four evaluation metrics, namely ADE, FDE, RMSE, and MR values; in terms of prediction real-time, the inference speed of SDT-ATT is 0.081ms. Therefore, the vehicle trajectory prediction model SDT-ATT meets the accuracy and real-time requirements for practical use.
- This research models the multivariate interactions (temporal, spatial and dynamic interactions) of the vehicle based on the multi-headed attention mechanism, and demonstrates the necessity of each interaction module by constructing ablation studies. The data from the ablation studies show that the absence of any interaction module will lead to worse prediction results, where the increase in prediction error due to the absence of temporal and spatial interactions is about the same and less than the increase in prediction error due to the absence of kinematic interactions. This indicates the necessity and importance of multivariate interaction modeling for vehicle trajectory prediction, and there are differences in the importance of each interaction.

The vehicle trajectory prediction method proposed in this paper can effectively model the multivariate interaction between the predicted vehicle and the neighbor vehicles, and the model performance meets the requirements of accuracy and real-time, which is of some practical significance. Meanwhile, this research provides some theoretical support

and technical support for the multivariate interaction modeling of vehicle trajectory prediction, which contributes to the research and development of intelligent and connected vehicles. In the future, more in-depth research will be carried out on the multivariate interaction of vehicles. Meanwhile, the popularity of diffusion models [42], [43] in the field of trajectory prediction has aroused our interest. We will try to introduce diffusion models in subsequent trajectory prediction work in order to obtain better model performance.

REFERENCES

- [1] K. Li, X. Chang, J. Li, Q. Xu, B. Gao, and J. Pan, "Cloud control system for intelligent and connected vehicles and its application," *Automot. Eng.*, vol. 42, no. 12, pp. 1595–1605, Dec. 2020.
- [2] C. X. Li, S. B. Lu, B. H. Zhang, W. J. Wu, and J. F. Lu, "Human vehicle steering collision avoidance path planning based on pedestrian location prediction," *Automot. Eng.*, vol. 43, no. 6, pp. 877–884, Jun. 2021.
- [3] M. Bahrman, C. Hubmann, A. Lawitzky, M. Aeberhard, and D. Wollherr, "A combined model- and learning-based framework for interaction-aware maneuver prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 6, pp. 1538–1550, Jun. 2016.
- [4] H. Zhao, J. Gao, T. Lan, C. Sun, B. Sapp, B. Varadarajan, Y. Shen, Y. Shen, Y. Chai, C. Schmid, C. Li, and D. Anguelov, "TNT: Target-driven trajectory prediction," in *Proc. Conf. Robot Learn. (CoRL)*, London, U.K., Oct. 2021, pp. 895–904.
- [5] Y. Huang, J. Du, Z. Yang, Z. Zhou, L. Zhang, and H. Chen, "A survey on trajectory-prediction methods for autonomous driving," *IEEE Trans. Intell. Vehicles*, vol. 7, no. 3, pp. 652–674, Sep. 2022.
- [6] S. Ammoun and F. Nashashibi, "Real time trajectory prediction for collision risk estimation between vehicles," in *Proc. IEEE 5th Int. Conf. Intell. Comput. Commun. Process.*, Cluj-Napoca, Romania, Aug. 2009, pp. 417–422.
- [7] N. Brouwer, H. Kloeden, and C. Stiller, "Comparison and evaluation of pedestrian motion models for vehicle safety systems," in *Proc. IEEE 19th Int. Conf. Intell. Transp. Syst. (ITSC)*, Cologne, Germany, Nov. 2016, pp. 1–6.
- [8] B. Jin, B. Jiu, T. Su, H. Liu, and G. Liu, "Switched Kalman filter-interacting multiple model algorithm based on optimal autoregressive model for manoeuvring target tracking," *IET Radar, Sonar Navigat.*, vol. 9, no. 2, pp. 199–209, Feb. 2015.
- [9] M. Althoff and A. Mergel, "Comparison of Markov chain abstraction and Monte Carlo simulation for the safety assessment of autonomous cars," *IEEE Trans. Intell. Transp. Syst.*, vol. 12, no. 4, pp. 1237–1247, Dec. 2011.
- [10] Y. Guo, V. V. Kalidindi, M. Arief, W. Wang, J. Zhu, H. Peng, and D. Zhao, "Modeling multi-vehicle interaction scenarios using Gaussian random field," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, Auckland, New Zealand, Oct. 2019, pp. 3974–3980.
- [11] P. Kumar, M. Perrollaz, S. Lefèvre, and C. Laugier, "Learning-based approach for online lane change intention prediction," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Gold Coast, QLD, Australia, Jun. 2013, pp. 797–802.
- [12] Q. Deng and D. Söfker, "Improved driving behaviors prediction based on fuzzy logic-hidden Markov model (FL-HMM)," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2018, pp. 2003–2008.
- [13] S. Qiao, D. Shen, X. Wang, N. Han, and W. Zhu, "A self-adaptive parameter selection trajectory prediction approach via hidden Markov models," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 1, pp. 284–296, Feb. 2015.
- [14] B. R. Kiran, I. Sobh, V. Talpaert, P. Mannion, A. A. A. Sallab, S. Yogamani, and P. Pérez, "Deep reinforcement learning for autonomous driving: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 4909–4926, Jun. 2022.
- [15] T. Fernando, S. Denman, S. Sridharan, and C. Fookes, "Deep inverse reinforcement learning for behavior prediction in autonomous driving: Accurate forecasts of vehicle motion," *IEEE Signal Process. Mag.*, vol. 38, no. 1, pp. 87–96, Jan. 2021.
- [16] K. Zhang, X. Feng, L. Wu, and Z. He, "Trajectory prediction for autonomous driving using spatial-temporal graph attention transformer," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22343–22353, Nov. 2022.

- [17] K. Shi, Y. Wu, H. Shi, Y. Zhou, and B. Ran, "An integrated car-following and lane changing vehicle trajectory prediction algorithm based on a deep neural network," *Phys. A, Stat. Mech. Appl.*, vol. 599, Aug. 2022, Art. no. 127303.
- [18] D. Ma, X. Song, and P. Li, "Daily traffic flow forecasting through a contextual convolutional recurrent neural network modeling inter- and intra-day traffic patterns," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 5, pp. 2627–2636, May 2021.
- [19] X. Ji, C. Fei, X. He, Y. Liu, and Y. Liu, "Intention recognition and trajectory prediction for vehicles using LSTM network," *China J. Highway Transp.*, vol. 32, no. 6, pp. 34–42, Jun. 2019.
- [20] K. Min, D. Kim, J. Park, and K. Huh, "RNN-based path prediction of obstacle vehicles with deep ensemble," *IEEE Trans. Veh. Technol.*, vol. 68, no. 10, pp. 10252–10256, Oct. 2019.
- [21] N. Deo and M. M. Trivedi, "Convolutional social pooling for vehicle trajectory prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Salt Lake City, UT, USA, Jun. 2018, pp. 1549–15498.
- [22] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human trajectory prediction in crowded spaces," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 961–971.
- [23] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, Dec. 2017, pp. 5998–6008.
- [24] K. Messaoud, I. Yahiaoui, A. Verroust-Blondet, and F. Nashashibi, "Attention based vehicle trajectory prediction," *IEEE Trans. Intell. Vehicles*, vol. 6, no. 1, pp. 175–185, Mar. 2021.
- [25] X. Chen, H. Zhang, F. Zhao, Y. Hu, C. Tan, and J. Yang, "Intention-aware vehicle trajectory prediction based on spatial-temporal dynamic attention network for Internet of Vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 19471–19483, Oct. 2022.
- [26] H. Guo, Q. Meng, D. Cao, H. Chen, J. Liu, and B. Shang, "Vehicle trajectory prediction method coupled with ego vehicle motion trend under dual attention mechanism," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–16, 2022.
- [27] Y. Wu, G. Chen, Z. Li, L. Zhang, L. Xiong, Z. Liu, and A. Knoll, "HSTA: A hierarchical spatio-temporal attention model for trajectory prediction," *IEEE Trans. Veh. Technol.*, vol. 70, no. 11, pp. 11295–11307, Nov. 2021.
- [28] L.-S. Jin, M. Gao, B.-C. Guo, X.-Y. Xie, and S.-R. Zhang, "Driver perspective trajectory prediction based on spatiotemporal fusion LSTM network," *China J. Highway Transp.*, vol. 35, no. 4, pp. 325–332, 2022.
- [29] L. I. Ke-Qiang, X. Hui, and L. I. U. Jin-Xin, "Multiple object motion trajectory prediction for vulnerable road user," *China J. Highway Transp.*, vol. 35, no. 1, pp. 298–315, 2022.
- [30] Z. Sheng, Y. Xu, S. Xue, and D. Li, "Graph-based spatial-temporal convolutional network for vehicle trajectory prediction in autonomous driving," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 17654–17665, Oct. 2022.
- [31] L. Xin, P. Wang, C.-Y. Chan, J. Chen, S. E. Li, and B. Cheng, "Intention-aware long horizon trajectory prediction of surrounding vehicles using dual LSTM networks," in *Proc. 21st Int. Conf. Intell. Transp. Syst. (ITSC)*, Nov. 2018, pp. 1441–1446.
- [32] Y. Su, J. Du, Y. Li, X. Li, R. Liang, Z. Hua, and J. Zhou, "Trajectory forecasting based on prior-aware directed graph convolutional neural network," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 9, pp. 16773–16785, Sep. 2022.
- [33] J. Guo, Z. He, Y. Luo, and K. Li, "Vehicle cut-in trajectory prediction based on deep learning in a human-machine mixed driving environment," *Automot. Eng.*, vol. 44, no. 2, pp. 153–160, 2022.
- [34] L. Shi, L. Wang, C. Long, S. Zhou, M. Zhou, Z. Niu, and G. Hua, "SGCN: Sparse graph convolution network for pedestrian trajectory prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Nashville, TN, USA, Jun. 2021, pp. 8990–8999.
- [35] K. Gao, X.-H. Li, L. Hu, B. Chen, and R.-H. Du, "Lane change intention prediction of CNN-LSTM based on multi-head attention," *J. Mech. Eng.*, vol. 58, no. 22, pp. 369–378, 2022.
- [36] L. Hou, L. Xin, S. E. Li, B. Cheng, and W. Wang, "Interactive trajectory prediction of surrounding road users for autonomous driving using structural-LSTM network," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 11, pp. 4615–4625, Nov. 2020.
- [37] X. Li, X. Ying, and M. C. Chuah, "GRIP: graph-based interaction-aware trajectory prediction," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, Auckland, New Zealand, Oct. 2019, pp. 3960–3966.
- [38] T. Zhang, W. Song, M. Fu, Y. Yang, and M. Wang, "Vehicle motion prediction at intersections based on the turning intention and prior trajectories model," *IEEE/CAA J. Autom. Sinica*, vol. 8, no. 10, pp. 1657–1666, Oct. 2021.
- [39] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.
- [40] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Inform: Beyond efficient transformer for long sequence time-series forecasting," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 12, pp. 11106–11115.
- [41] *Traffic Analysis Tools: Next Generation Simulation-FHWA Operations*. Accessed: Nov. 24, 2020. [Online]. Available: <https://ops.fhwa.dot.gov/trafficanalysistools/ngsim.htm>
- [42] W. Mao, C. Xu, Q. Zhu, S. Chen, and Y. Wang, "Leapfrog diffusion model for stochastic trajectory prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada, Jun. 2023, pp. 5517–5526.
- [43] Y. Liu, Z. Ye, B. Li, and L. Yao, "Uncertainty-aware pedestrian trajectory prediction via distributional diffusion," Mar. 2023, *arXiv:2303.08367*.



DONGXIAN SUN received the B.S. degree in vehicle engineering from Liaocheng University, Liaocheng, China, in 2018, and the M.S. degree in vehicle operation engineering from Jilin University, Changchun, China, in 2021. He is currently pursuing the Ph.D. degree in mechanical engineering with the School of Mechanical Engineering, Beijing Institute of Technology, Beijing, China. His research interests include vehicle trajectory prediction, driving risk assessment, and driver behavior analysis.



HONGWEI GUO received the B.E. degree in traffic engineering and the M.E. degree in transportation engineering from the Beijing Institute of Technology and the Ph.D. degree in transportation planning and management from Beijing Jiaotong University. He is currently an Associate Professor with the School of Mechanical Engineering, Beijing Institute of Technology. His research interests include traffic behavior analysis and traffic flow simulation.



WUHONG WANG received the Ph.D. degree from Southwest Jiaotong University. He spent two years as a Postdoctoral Fellow with Beihang University, from 1997 to 1999. Then, he has been with the Beijing Institute of Technology as an Associate Professor, since 1999, and then a Full Professor. He stayed with the Computer Vision Laboratory, Institute of Industrial Science, The University of Tokyo, Japan, as a JSPS Research Fellow, in 2002 and 2004, and a Visiting Professor, in 2005. From 2005 to 2007, he was with Lehrstuhl für Ergonomie, Technische Universität München, serviced as a Humboldt Research Fellow. He conducted research with the Transportation Research Institute, Hasselt University, Belgium, in 2014. His research interests include intelligent connected vehicles and assisted driving, green transportation behavior and safety, human factors, and human-machine interaction.

...